

PROJECT REPORT : DIGITAL **THERMOMETER USING 8051** **MICROCONTROLLER**

Tools used:

- **Simulation Software: Proteus**
- **Compiler: Keil uVision**

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INTRODUCTION

This project implements a digital thermometer using an 8051 microcontroller. The LM35 temperature sensor is used to measure temperature, but since the 8051 lacks an inbuilt ADC, an external ADC0804 is used to convert the analog output of LM35 into a digital value. The microcontroller processes this data and displays the temperature reading on a 16x2 LCD display. This system can be used for weather monitoring, home automation, or industrial applications.

OBJECTIVES

- **Accurate Temperature Measurement:** Use a temperature sensor -LM35 to measure the ambient temperature.
- **Signal Conversion and Processing:** Utilize the ADC (Analog-to-Digital Converter) to convert the analog temperature signal into a digital format.
- **Microcontroller Programming:** Develop embedded C code to process the digital temperature data using the 8051 microcontroller.
- **Display Interface:** Display the measured temperature in degrees Celsius (°C) or Fahrenheit (°F) on an LCD module.
- **User-Friendly Output:** Provide a clear, real-time temperature output with appropriate precision.

COMPONENTS USED

Microcontroller: 8051

IDE: Keil μ Vision

Sensors/Active Devices:

Custom PCB: No

- Temperature Sensor (LM35)
- LCD Display (16x2)
- Analog-to-Digital Converter (ADC): ADC0804

Trainer Board: 8051 Trainer Kit

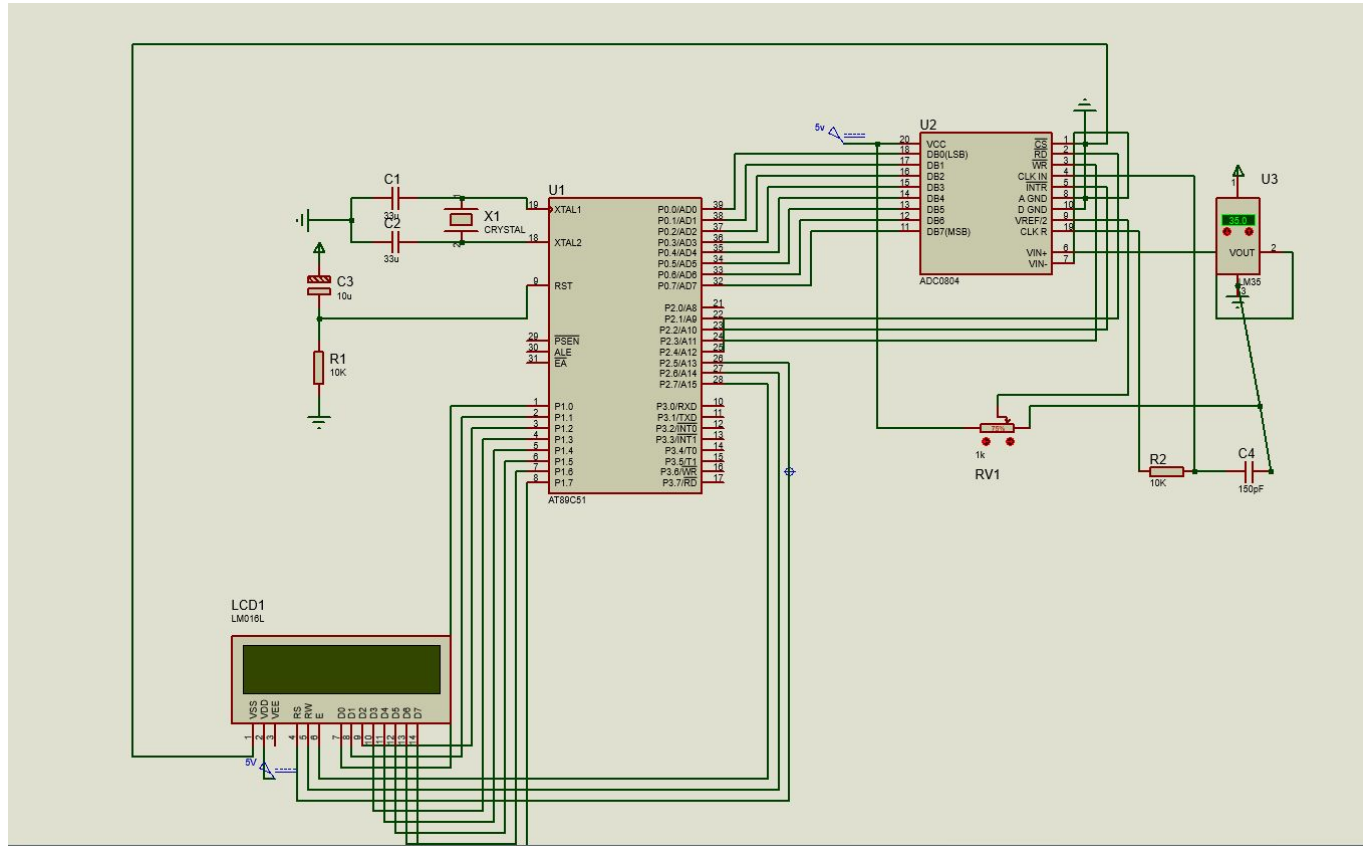
Passive Components:

- Resistors: 1k Ω , 10k Ω
- Capacitors: 100nF, 10 μ F

RAM/ROM Capacity and Other Peripherals:

- RAM: 128 bytes
- ROM: 4KB
- I/O Ports: 32

CIRCUIT DIAGRAM



WORKING PRINCIPLE

Temperature Sensing:

- The LM35 temperature sensor measures the ambient temperature and produces an analog voltage proportional to the temperature (e.g., 10 mV/°C).

Analog-to-Digital Conversion:

- The analog voltage output from the LM35 is fed into the ADC0804, which converts the analog signal into a digital value.

Data Processing:

- The 8051 microcontroller reads the digital value from the ADC0804 and processes it to convert the data into a temperature reading using appropriate scaling formulas.

Display Output:

- The processed temperature value is sent to a 16x2 LCD display, which shows the temperature in degrees Celsius in real time.

Continuous Monitoring:

- The system operates in a continuous loop, repeatedly reading the sensor, converting the data, and updating the display to provide real-time temperature readings.

CODE

The system is programmed in Keil uVision in embedded C.

ADVANTAGES

- The LM35 sensor provides accurate temperature readings with a linear output (10 mV per °C), reducing errors.
- The system uses affordable and widely available components like the 8051 microcontroller, LM35 sensor, ADC0804, and LCD display, thus cost-effective design.
- The system uses affordable and widely available components like the 8051 microcontroller, LM35 sensor, ADC0804, and LCD display.
- Unlike mercury thermometers, this digital solution eliminates the risk of mercury exposure, making it safer for users.

APPLICATION

- Used to measure ambient temperature for local weather updates and meteorological applications.
- Integrated into smart home systems for monitoring and maintaining room temperature.
- Used in greenhouses and storage areas to monitor and control temperature for optimal crop growth and food preservation.
- Can be adapted for medical thermometers to measure body temperature accurately and safely.

CONCLUSION

The digital thermometer using the 8051 microcontroller is a simple, cost-effective, and efficient system for real-time temperature measurement. By interfacing the LM35 sensor, ADC0804, and a 16x2 LCD, the microcontroller successfully processes and displays accurate temperature readings. This project demonstrates the practical use of microcontrollers in embedded systems and highlights its potential applications in weather monitoring, home automation, and industrial temperature control systems. It also serves as a foundation for future enhancements, such as adding wireless communication, data logging, or threshold-based alerts.